Autoscaling Group [ASG]

Autoscaling Group + ELB = Highest availability [Superpower]

↑

Based on Cloud watch Metric [Such as CPU utilisation or Custom metric] & Alarm

→ Goal of ASG:

1] Scale-out → add instances to match increase load

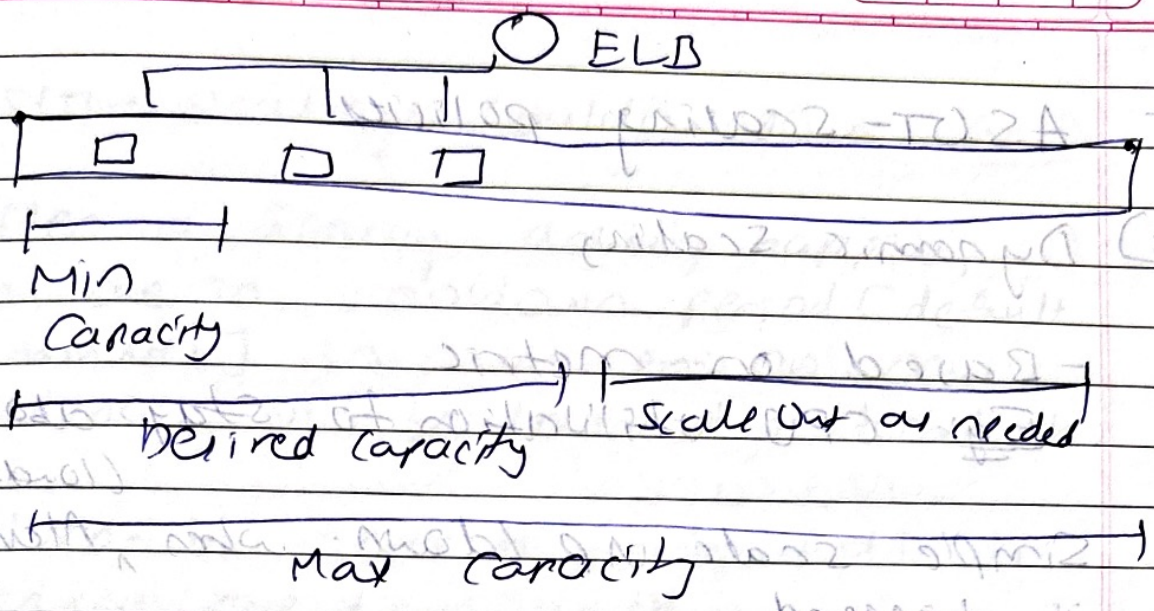
2] Scale-in → remove instance to match decreased load.

3] Ensure min & max instance are running

4] Automatically register new instances to ELB

5] Recreate if previous one got deleted

→ ASG is free, you get paid for resources created from it.



→ What ASG needs :-

1] A Launch template
2] ELB information
3] Network + Subnet info
4] Min / Max / Intial size capacity
5] Scaling policies

ASG - scaling policies

1.] Dynamic scaling

- Based on metric
Eg: CPU utilization to stay above 40%
(Cloudwatch Alarm)

Simple scale up/down when Alarm is triggered.

2.] Schedule Scaling

Scale out at time period when you know CPU utilization etc will increase

3.] Predictive scaling [Machine Learning driven]

Analyze historical Load → Form pattern & forecast → schedule scaling action.
[Planning ahead]

Metric to scale on:-

1.] CPU - Utilization → Average CPU utilization

2.] Request Count PerTarget → To make sure the number of request per EC2 is stable.

3.] Average Network In/Out [if your application is network bound]

4.] Any Custom metrics

ASG - Scaling cooldown

→ After a scaling activity happens, you are in cooldown period [default 5 minutes] in which ~~on~~ no new instance will be launched or terminated.

→ Advice - Use ready made AMI to reduce cooldown period.

Dynamic scaling policy

- Simple Scaling
  - Based on metric, scale in / out
- Step scaling
  - If metric is this then scale out
  - If this then scale down
  - ie Insert
- Target Tracking Policy
  - Based on a target given to metric
  - ie ~~Increase utilization 40%~~ Scale out.
  - Maintain CPU utilization at average 40%.